

Statistical and Computational Models

Lecture 7: Binary outcomes

Daniel de Kadt

Department of Government, Cornell University · Fall 2026

Roadmap

Modelling a probability

Logit and probit

Fitting the logit model

Binary outcomes

Very often we encounter outcomes of the form:

$$Y_i \in \{0, 1\}$$

Examples in political science: **many**

- Turnout
- Roll-call votes
- Vote choice (more later)
- Interstate conflict
- Alliances
- Regime type (1 = democracy)
- etc. etc.

Binary outcomes

Our running example is regime type (**1** = democracy, **0** = autocracy).

So far we have modeled this Y_i with a single parameter, π .

Let's add two covariates to the simulation: **lgdp** and **oil**.

y	lgdp	oil
1	9.77	1
1	9.23	0
1	8.32	0
0	6.49	1

Target: the **conditional** probability $\mathbb{E}[Y_i | \mathbf{x}_i] = \mathbb{P}(Y_i = 1 | \mathbf{x}_i)$.

The linear probability model

Goal: learn how $\mathbb{P}(Y_i = 1 \mid \mathbf{x}_i)$ changes with $\mathbf{x}_i$.

Many models, $\mathbb{P}(Y_i = 1 \mid \mathbf{x}_i) = g^{-1}(\mathbf{x}_i^\top \boldsymbol{\beta})$, depending on the choice of g .

One option, g^{-1} the identity, is the **linear probability model** (LPM):

$$\mathbb{P}(Y_i = 1 \mid \mathbf{x}_i) = \mathbf{x}_i^\top \boldsymbol{\beta}, \quad \text{or equivalently} \quad Y_i \sim \text{Bernoulli}(\pi_i), \quad \pi_i = \mathbf{x}_i^\top \boldsymbol{\beta}$$

Read: the probability that $Y_i = 1$ comes from a weighted sum of the covariates $\mathbf{x}_i$.

Easy to fit, and β_j reads as a change in probability:

```
lpm <- lm(y ~ oil + lgdp, data = countries)
```

$\hat{\beta}_{\text{oil}} = -0.22$: at the same `lgdp`, oil states are 22 percentage points less likely to be democracies.

The linear probability model

Three downsides:

1. With continuous $\mathbf{x}$, a constant β_j **must be wrong**: probabilities must flatten near 0 and 1.
2. So predicted 'probabilities' for units can fall **outside** $[0, 1]$: in our case, 27 of 205 do so, from -0.413 to 1.311 .
3. The errors in $Y_i = \mathbf{x}_i^\top \beta + \varepsilon_i$ are **always heteroskedastic**. Why?

$$\text{Var}[\varepsilon_i | \mathbf{x}_i] = \text{Var}[Y_i | \mathbf{x}_i] = \mathbf{x}_i^\top \beta (1 - \mathbf{x}_i^\top \beta)$$

Read: the variance $\pi_i(1 - \pi_i)$ moves with $\mathbf{x}_i$.

Guidance for the LPM

Despite these downsides, the simplicity of the LPM means it is very popular.

Some guidance:

1. For **average effects** with one binary, ignorable x , the model is **saturated**, so β is exactly the ATE.
2. Generally, LPM slopes are often quite similar to alternative models; it is a good idea to **check and report alternative models** even if you use an LPM.
3. Always use **heteroskedasticity-robust** standard errors, `{sandwich}::vcovHC()` or `{estimatr}::lm_robust()`.

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Keeping $\hat{Y}$ in $[0,1]$

Goal: $0 \leq \mathbb{P}(Y_i = 1 \mid \mathbf{x}_i) \leq 1$ for any $\mathbf{x}_i$

But $\mathbf{x}_i^\top \boldsymbol{\beta}$ can be **any real number**. We need a g^{-1} that maps $\mathbb{R} \rightarrow [0, 1]$, and is increasing such that a larger $\mathbf{x}_i^\top \boldsymbol{\beta}$ gives a larger probability.

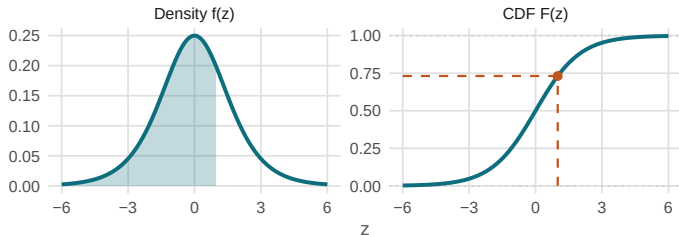
We already know a class of functions that does exactly this!

The **CDF**: defined for every $\mathbf{z} \in \mathbb{R}$, between 0 and 1, never decreasing.

Aside: the CDF

For a random variable Z drawn from a probability distribution, the **cumulative distribution function** (CDF) is

$$F(z) = \mathbb{P}(Z \leq z) = \int_{-\infty}^z f(t) dt$$



Example: the standard **logistic**. Shaded area left of $z = 1$: $F(1) = 0.73$.

Keeping $\hat{Y}$ in $[0,1]$

Solution: make g^{-1} a **CDF**, F :

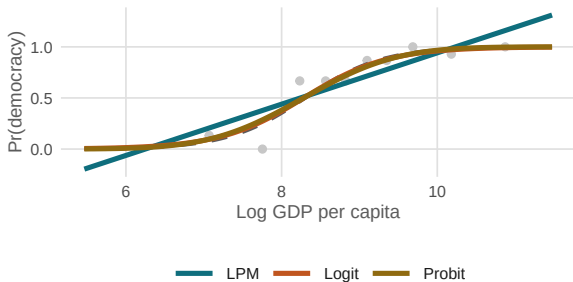
$$\mathbb{P}(Y_i = 1 \mid \mathbf{x}_i) = g^{-1}(\mathbf{x}_i^\top \beta) = F(\mathbf{x}_i^\top \beta), \quad \text{logit}^{-1}(z) = \frac{e^z}{1 + e^z}$$

Logit: logistic CDF, $F = \text{logit}^{-1}$ (a.k.a. **inverse logit**).

Probit: standard normal CDF, $F = \Phi$.

Note: F is **not** the CDF of Y_i , which is Bernoulli (using a CDF is just a convenient way to ensure $0 \leq \pi_i \leq 1$).

Keeping $\hat{Y}$ in $[0,1]$



Grey dots: binned means at $\Delta \log = 0.1$. Dashed: the true DGP.

LPM underestimates the change in $\Pr(\text{democracy})$ near the center, overshoots at both ends.

Logit, and probit: little practical difference, but note that $\hat{\beta}$ are completely different between the models.

A generalized linear model

Recall: $Y_i \sim \text{Bernoulli}(\pi_i)$. And we want π_i to be a function of $\mathbf{x}_i$:

1. A distribution from the exponential family (Bernoulli, normal, ...).
2. A linear predictor: $\eta_i = \mathbf{x}_i^\top \boldsymbol{\beta}$.
3. A link: $g(\mathbb{E}[Y_i | \mathbf{x}_i]) = \eta_i$.

g is the **link**. We have been working with its inverse, $\mathbb{E}[Y_i | \mathbf{x}_i] = g^{-1}(\eta_i)$.

Logit: $g(\pi_i) = \log\{\pi_i/(1 - \pi_i)\}$, so $g^{-1} = \text{logit}^{-1}$.

This is a **generalized linear model**.

Using the link function

Recall the Bernoulli for one country: $\pi_i^{y_i}(1 - \pi_i)^{1-y_i}$.

Substitute in the inverse logit link, $\pi_i = e^{\eta_i} / (1 + e^{\eta_i})$, and take the log:

$$\log \pi_i = \eta_i - \log(1 + e^{\eta_i}), \quad \log(1 - \pi_i) = -\log(1 + e^{\eta_i})$$

$$\begin{aligned}\ell_i &= y_i \log \pi_i + (1 - y_i) \log(1 - \pi_i) \\ &= y_i \left[\eta_i - \log(1 + e^{\eta_i}) \right] - (1 - y_i) \log(1 + e^{\eta_i}) \\ &= y_i \eta_i - \underbrace{\left[y_i + (1 - y_i) \right]}_{= 1} \log(1 + e^{\eta_i}) \\ &= y_i \eta_i - \log(1 + e^{\eta_i})\end{aligned}$$

y_i now multiplies η_i only, so $\partial \ell_i / \partial \eta_i = y_i - \pi_i$: the **residual**.

Links and distributions

Under the logit, y_i multiplies $\eta_i = \log\{\pi_i/(1 - \pi_i)\}$. This quantity is called the **log odds**.

Every **exponential-family** distribution has one such quantity, its **canonical parameter** θ : whatever its log-likelihood multiplies by $\mathbf{y}$, so that $\partial \ell_i / \partial \theta_i = y_i - \mu_i$.

The **canonical link** sets $\theta_i = \eta_i$, so $\mathbf{g}$ comes from the choice of distribution for Y_j :

Y_j	Mean	θ	Canonical $\mathbf{g}$	Model
Normal	μ	μ	identity	OLS
Bernoulli	π	$\log\{\pi/(1 - \pi)\}$	logit	logit

This choice is a convenience (e.g. probit uses $\mathbf{g} = \Phi^{-1}$).

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The likelihood, with covariates

Assume that Y_i is drawn from a Bernoulli, with π_i depending on covariates $\mathbf{x}_i$ through parameters β (that we care about).

Write down the log-likelihood given your assumption:

$$L(\beta; \mathbf{Y}, \mathbf{X}) = \prod_{i=1}^n \pi_i^{y_i} (1 - \pi_i)^{1-y_i}, \quad \pi_i = \text{logit}^{-1}(\mathbf{x}_i^\top \beta)$$
$$\ell(\beta) = \sum_{i=1}^n \left\{ y_i \log \pi_i + (1 - y_i) \log(1 - \pi_i) \right\}$$

Read: ℓ is large when π_i is near 1 for democracies, near 0 for autocracies.

Guess β : each guess gives every country its own π_i (based on $\mathbf{x}_i$).

At $\hat{\beta}$, $\ell = -86.72$. For our no-covariate $\hat{\pi}$, it was -138.74 .

The likelihood, with covariates

Recall the **score**, the derivative of ℓ in β (after tedious math):

$$\mathbf{s}(\beta) = \sum_{i=1}^n (y_i - \pi_i) \mathbf{x}_i$$

Set $\mathbf{s}(\beta)$ to zero and solve for $\hat{\beta}$. With only an intercept, $\sum_i (y_i - \pi) = 0$ gives the mean-only $\hat{\pi} = r/n$.

This looks a lot like the OLS **normal equations**, with $\hat{\pi}_i$ where $\mathbf{x}_i^\top \hat{\beta}$ was:

$$\text{logit: } \sum_{i=1}^n (y_i - \hat{\pi}_i) \mathbf{x}_i = \mathbf{0}$$

$$\text{OLS: } \sum_{i=1}^n (y_i - \mathbf{x}_i^\top \hat{\beta}) \mathbf{x}_i = \mathbf{0} \quad \Longleftrightarrow \quad \mathbf{X}^\top \mathbf{X} \hat{\beta} = \mathbf{X}^\top \mathbf{y}$$

But π_i is **nonlinear** in β : no closed form. We use the Newton–Raphson algorithm (`glm()`: Fisher scoring in R).

Fitting it in R

Fitting the model in R using `glm()` gives us:

```
logit <- glm(y ~ oil + lgdp, data = countries, family = binomial,  
             control = glm.control(epsilon = 1e-10))  
round(summary(logit)$coefficients[, 1:2], 3)
```

	Estimate	Std. Error
(Intercept)	-15.329	2.343
oil	-1.560	0.429
lgdp	1.851	0.273

Note: `family = binomial` with a $\{0, 1\}$ outcome. This states that $Y_i \sim \text{Bernoulli}$ (one trial), and imposes the logit link by default.

The actual DGP had $\beta = (-16.6, -1.5, 2.0)^\top$, so the logit has done well!

References

- Rodríguez, G. *Generalized Linear Models*, Princeton lecture notes – **3.1–3.3, 3.7**, 3.4–3.6, App. B. grodriguez.github.io/glms.